

HW 2: Significance Test for Means

_____ 1) Sadie is thinking about quitting her current job and take an opportunity that allows her start her business working from home. In order for her business to be profitable, she has to have an average monthly revenue of \$4,500. Anything less than that, and she risks losing her business. Sadie is going to select a random sample of 10 small businesses like hers and ask them their monthly income. She will test the hypotheses: $H_0: \mu = \$4,500$ versus $H_a: \mu > \$4,500$. Describe a Type I error and its consequence.

- (A) The consequences of a Type I error cannot be determined without an α -level.
- (B) You believe the average revenues of small businesses like yours is more than \$4,500, so you open your business and have a high risk of losing your business.
- (C) You believe the average revenues of small businesses like yours is more than \$4,500, so you open your business and have a high probability of a successful business.
- (D) You believe the average revenues of small businesses like yours is \$4,500 or less, so you do not open your business and will miss out on an opportunity to own a successful business.
- (E) You believe the average revenues of small businesses like yours is \$4,500 or less, so you do not open your business because other small businesses like yours are not successful.

_____ 2) When are you able to use a confidence interval to investigate a claim instead of computing a p-value in a significance test?

- (A) A confidence interval can be created to test any claim in a significance test.
- (B) In any significance test with a two-sided alternative hypothesis.
- (C) Only when the hypothesized value of the parameter is NOT in the confidence interval
- (D) Only when you are conducting a significance test with a one-sided alternative
- (E) Only when doing a test for a population mean

_____ 3) An appropriate 95% confidence interval for μ has been calculated as $(-0.55, 1.25)$ based on $n = 20$ observations from a population with a normal distribution. If we wish to use this confidence interval to test the hypothesis $H_0: \mu = 0$ against $H_a: \mu \neq 0$, which of the following is a legitimate conclusion?

- (A) Reject H_0 at the $\alpha = 0.05$ level of significance.
- (B) Fail to reject H_0 at the $\alpha = 0.05$ level of significance.
- (C) Reject H_0 at the $\alpha = 0.10$ level of significance.
- (D) Fail to reject H_0 at the $\alpha = 0.10$ level of significance.
- (E) We cannot perform the required test since we do not know the value of the test statistic.

_____ 4) What is the average teacher salary in the United States? One district claims that it is \$43,500 and will therefore set the salary level based on that. The teacher's union says that it is incorrect and the actual income is higher and the district should set their salary level higher. The teacher's union selects an SRS of 15 teachers from around the US and finds that the mean salary is \$45,000 with a sample standard deviation of \$2,800. Assuming that the distribution of all teachers' salaries is approximately normally distributed, what is the value of the t-test statistic and the P-value for a test of the hypothesis $H_0: \mu = 43,500$ against $H_a: \mu > 43,500$?

- (A) $t = 2.075$ $0.01 < p < 0.05$
- (B) $t = 2.075$ $0.001 < p < 0.01$
- (C) $t = -2.075$ $0.01 < p < 0.05$
- (D) $t = -2.075$ $0.001 < p < 0.01$
- (E) $t = -2.075$ $p > 0.05$

_____ 5) A company that produces over-the-counter pain relief medication advertises that their current formula provides relief in, on average, 15 minutes. They believe that have developed a new formula that is expected to bring relief faster. If the research team wishes to test the new formula to see if it truly improves pain relief time, which of the following should be the appropriate null and alternative hypotheses, if μ = the mean time for the medication to bring pain relief?

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|---|--|
| (A) $H_0: \mu = 15$
$H_a: \mu \neq 15$ | (B) $H_0: \mu = 15$
$H_a: \mu < 15$ |
| (C) $H_0: \mu < 15$
$H_a: \mu = 15$ | (D) $H_0: \mu < 15$
$H_a: \mu > 15$ |
| (E) $H_0: \mu = 15$
$H_a: \mu > 15$ | |

_____ 6) A significance test was performed to test the null hypothesis $H_0: \mu = 1.5$ versus the alternative $H_a: \mu \neq 1.5$. A sample of size 15 produced a test statistic is $t = 2$. Assuming all conditions for inference were met, which of the following intervals contains the P-value for this test?

- (A) $p < 0.02$
- (B) $0.025 > p > 0.02$
- (C) $0.05 > p > 0.025$
- (D) $0.10 > p > 0.05$
- (E) $p > 0.10$

7) Functional Fitness claims that its new exercise program helps their members lose 5 pounds in the first month. To test this claim, you select a random sample of 13 members who completed the program and record their weight loss (in pounds) as follows:

5.5	5.2	6	5.5	6.8	4	4.2	5.5	6.2	5.1	4.8	6.6	3.9
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Is there evidence, at the 10% significance level, that this new exercise program helps members lose more than 5 pounds in their first month?